

1 Ideal Over-the-Air Computation Baseline

Before introducing the practical impairments inherent to wireless channels, such as Inter-Symbol Interference (ISI), multipath fading, and time synchronization errors, it is essential to establish a baseline model of Over-the-Air (OTA) computation under ideal conditions. The foundational premise of OTA computing is its ability to reduce the communication latency of data aggregation from the linear complexity shown in Eq. (1) to the constant complexity made possible by allowing K devices to transmit simultaneously [3].

$$\mathcal{O}(K) \longrightarrow \mathcal{O}(1) \quad (1)$$

In this idealized scenario, the wireless medium is assumed to be perfectly synchronized, lacking frequency-selective fading, and corrupted solely by Additive White Gaussian Noise (AWGN). This chapter aggregates the mathematical formulation, algorithmic MATLAB modeling, block-level Simulink architecture, and GNU Radio flowgraphs that demonstrate this ideal, unimpaired baseline. Establishing this benchmark is critical for quantifying the exact degradation caused by physical channel constraints in subsequent chapters.

1.a Mathematical Formulation of the Ideal Channel

Consider a distributed wireless sensor network consisting of K End Device (ED)s transmitting local analog values to a central Fusion Center (FC). Let $x_k(t)$ denote the continuous-time, real-valued signal transmitted by the k -th device. For analog OTA computation, uncoded Pulse Amplitude Modulation (PAM) is widely considered the optimal transmission scheme, as it linearly maps the sensor readings directly to the amplitude of the transmitted waveform [2], [4].

In an ideal Multiple Access Channel (MAC), we assume perfect Channel State Information (CSI) at the transmitters, which allows for flawless pre-equalization of phase shifts and path loss attenuation. Consequently, the electromagnetic waves superimpose perfectly at the receiving antenna of the FC. The received baseband signal $y(t)$ is mathematically expressed as the linear sum of the transmitted signals plus noise:

$$y(t) = \sum_{k=1}^K x_k(t) + n(t) \quad (2)$$

where the AWGN model is specified by Eq. (3). Because there is no multipath delay spread in this baseline model, the channel impulse response is effectively the Dirac delta function in Eq. (4). This implies that the current symbol is entirely unaffected by past or future symbols, resulting in zero ISI. To extract the arithmetic average of the K sensor readings, the FC simply applies a post-processing scaling factor of $1/K$ to the received aggregated signal $y(t)$.

$$n(t) \sim \mathcal{N}(0, \sigma^2) \quad (3)$$

$$h(t) = \delta(t) \quad (4)$$

1.b MATLAB Simulation and Code Analysis

To validate this superposition principle, the ideal environment can be modeled using MATLAB. The following script generates normally distributed simulated sensor readings, aligns them perfectly in time, and adds AWGN to emulate the ideal MAC.

Listing 1: MATLAB script for ideal OTA computation aggregation.

```

1      % Ideal OTA Computation Simulation
2      K = 3;                % Number of sensors
3      N = 1000;            % Number of samples
4      SNR_dB = 20;         % Signal-to-Noise Ratio
5
6      % Generate random temperature readings around 27 degrees
7      X = 27 + randn(K, N);
8
9      % The Air acts as an ideal adder (MAC Superposition)
10     Ideal_Sum = sum(X, 1);
11
12     % Add AWGN
13     Received_Signal = awgn(Ideal_Sum, SNR_dB, 'measured');
14
15     % Calculate the Fusion Center Average
16     Estimated_Average = Received_Signal / K;
17
18     % Display Mean Squared Error (MSE)
19     MSE = immse(Estimated_Average, mean(X, 1)); fprintf('Ideal Baseline MSE
        at %d dB SNR: %f\n', SNR_dB, MSE);

```

While Listing 1 demonstrates the fundamental principle in its unadulterated form, practical wireless systems necessitate pulse shaping to satisfy spectral constraints and facilitate signal recovery. Consequently, a more physically realistic model incorporates matched Raised Cosine (RC) filtering at both the transmitter and receiver. This architectural refinement introduces additional degrees of freedom—namely, the Symbols per Sample (sps) parameter and the roll-off factor (α)—which materially influence system performance. The enhanced implementation is presented in Listing 2.

Listing 2: MATLAB script for OTA computation aggregation using the Root Raised Cosine (RRC) filter.

```

1      txUpSamped = upsample(X', sps_val); % phase may need to be set
2
3      % Raised Cosine Filter
4      rcFilter = rcosdesign(aleph, span, sps_val, "sqrt");
5
6      rx_agg_signal = zeros(N * sps_val + length(rcFilter) - 1, 1);
7
8      for k = 1:K
9          rx_agg_signal = rx_agg_signal + conv(txUpSamped(:,k), rcFilter(:));
10     end
11
12     Received_Signal = awgn(rx_agg_signal, current_snr, 'measured');
13
14     matchedSignal = conv(Received_Signal, flipud(rcFilter(:)), 'same');
15
16     rxMatched = downsample(matchedSignal, sps_val);
17

```

```

18 start_idx = floor(span/2); stop_idx = floor(span/2) + mod(span,2) + 1;
19
20 % Truncate edges to avoid filter transient errors
21 Estimated_Average = rxMatched(start_idx:end-stop_idx, :) / K;
22
23 % Calculate and store MSE in the matrix
24 mse_results(snr_idx, a_idx) = immse(Estimated_Average', mean(X, 1));

```

As demonstrated by the script, when synchronization is perfect, the MSE is strictly a function of the AWGN variance. When however trying to implement a scenario that is closer to a real world usecase, it appears to be necessary to interject a filter pair like the raised cosine filter. In order for this to be accomplished, a matched root raised cosine filter shall be inserted on each end of channel and by doing that, more degrees of freedom shall be taken into account, like sps and Roll-off Factor (α). In diagram 1b the curve can be seen where in diagram 1a the ideal Mean Squared Error (MSE) can be shown.

Computational execution of the aforementioned algorithms reveals a critical distinction between the idealized and practical regimes. When temporal synchronization is mathematically perfect and no filtering is applied, the estimation accuracy is entirely governed by the noise floor; the MSE exhibits a direct proportionality to the AWGN variance and inverse quadratic scaling with the number of transmitting sensors. Mathematically, this relationship is expressed in Eq. (5), which demonstrates the fundamental advantage of cooperative multi-device aggregation [1], [3]. Conversely, when pulse shaping filters are incorporated into the transmission chain, the interplay between symbol timing parameters (sps) and spectral efficiency (α) introduces measurable deviation from the ideal theoretical bound.

$$\text{MSE} \propto \frac{\sigma^2}{K^2} \quad (5)$$

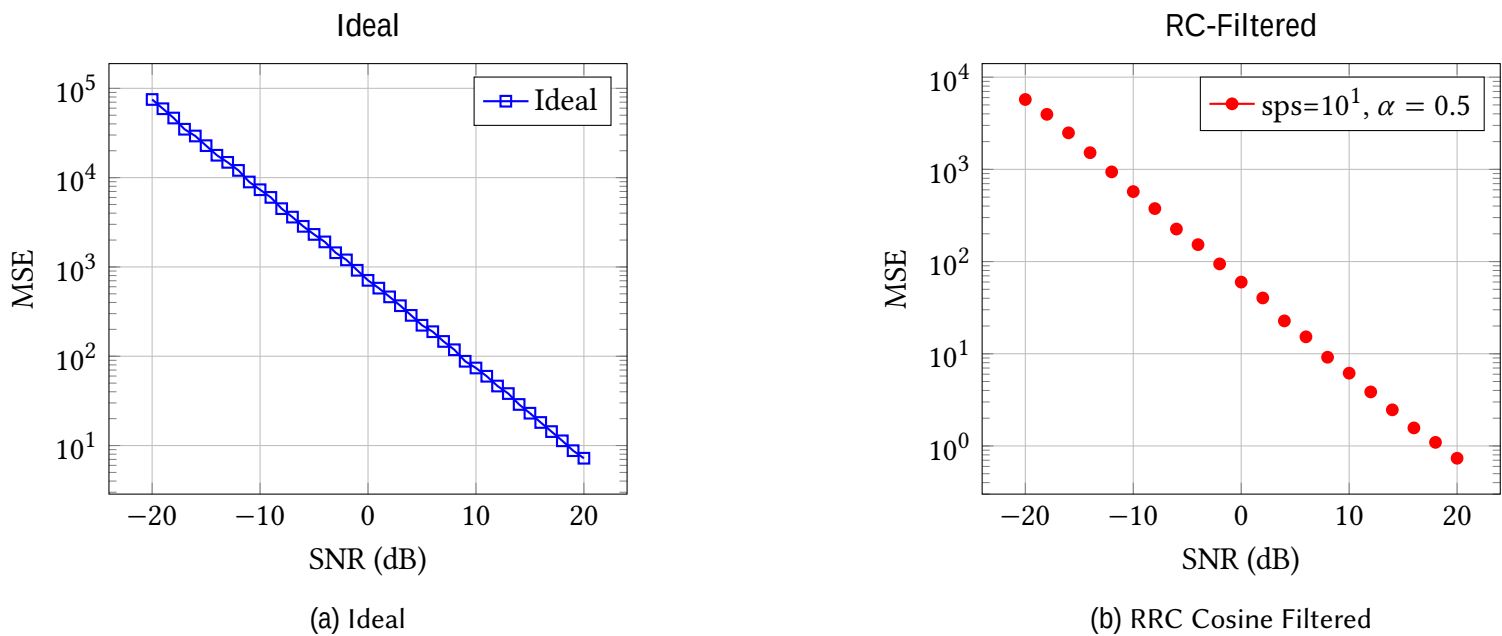


Figure 1: Comparison of MSE performance between the ideal baseline and the Raised Cosine filtered scenario.

Figure 1a presents the theoretical performance floor, wherein the MSE is a function solely of AWGN variance. In contrast, Figure 1b illustrates the performance envelope when matched Raised Cosine filters are deployed at both transmitter and receiver, with sps and α held constant. Despite the introduction of filtering, the channel remains free of multipath distortion and timing misalignment, ensuring that noise contributions remain the dominant performance-limiting factor.

A particularly noteworthy observation emerges from parametric analysis of the animation in Figure 2: while increasing sps generally yields marginal improvements in MSE performance across the SNR range examined, this enhancement

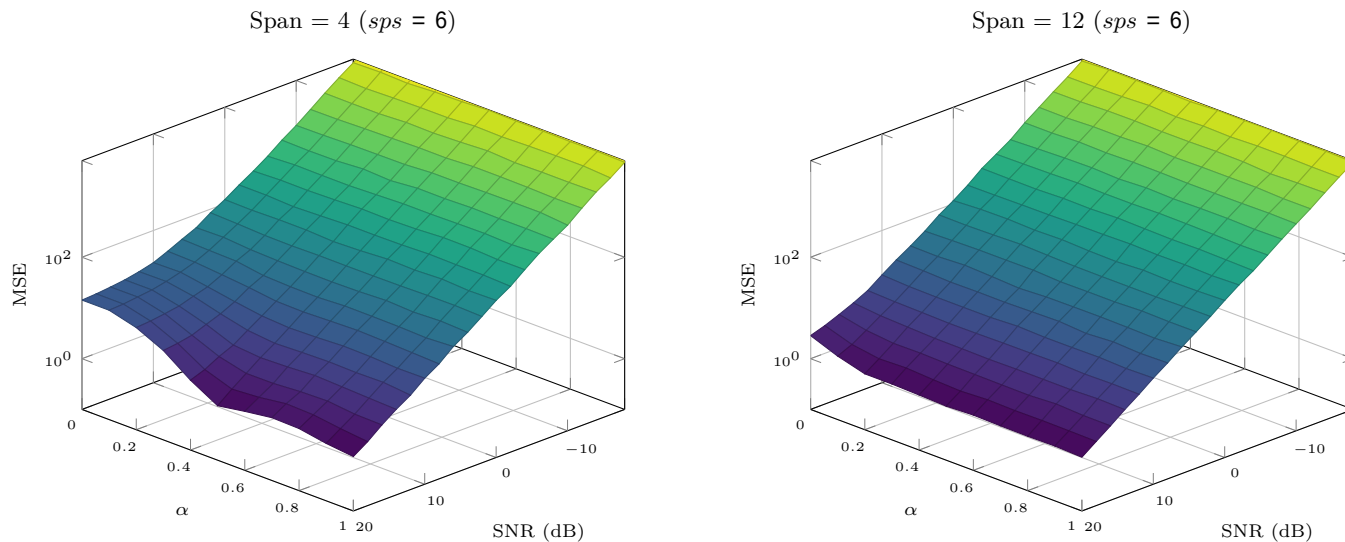
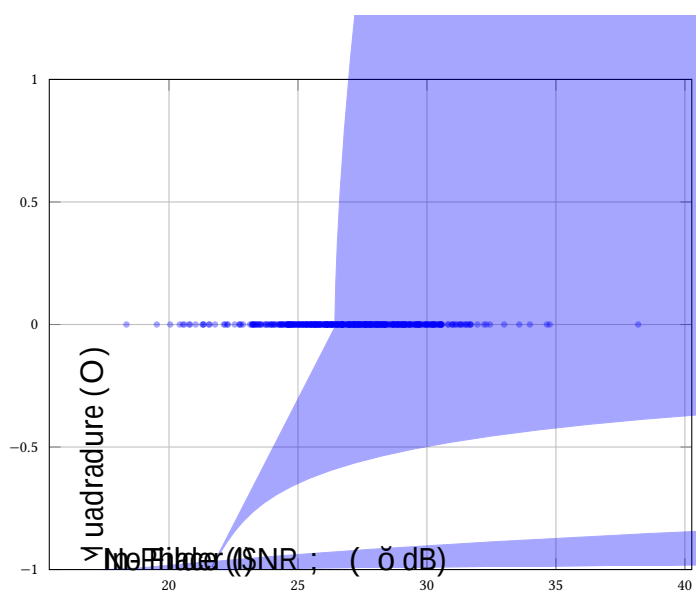


Figure 2: Synchronous 3D animation comparing the Over-The-Air MSE surface across sps scaling for filter span 4 (left) versus span 12 (right).

asymptotically approaches a limit and does not constitute convergence toward the theoretical ideal baseline (absent pulse shaping). This behavior underscores a fundamental trade-off inherent to practical pulse-shaping implementations: enhanced spectral efficiency through reduced sps incurs amplified inter-symbol interference, whereas elevated oversampling ratios improve timing robustness at the cost of increased computational complexity and memory consumption.

The eye and constellation diagrams provide valuable insights into the behavior of the ideal OTA system under different SPS configurations. As the number of samples per symbol increases, the eye opening widens, indicating reduced inter-symbol interference and improved signal integrity. Similarly, the constellation points become more distinct and well-separated, reflecting better performance in terms of error probability and overall system reliability.

The span parameter controls the width of the pulse shaping filter, and a larger span results in a more gradual transition band, leading to a fatter MSE curve across different roll-off factors.



(a) Unfiltered channel baseline.



(b) Matched filter dynamic response across roll-off α .

Figure 3: Constellation diagram comparison at SNR = 20 dB. Left: Unfiltered AWGN channel. Right: Animation demonstrating the suppression of truncation-induced ISI as α scales from 0.00 to 1.00 (span = 12, sps = 60).

The constellation diagrams in Figure 3 provide a visual representation of the received signal's integrity under both unfiltered and filtered conditions. The left subfigure illustrates the dispersion of constellation points in the absence of filtering,

indicating potential symbol overlap and increased error probability. In contrast, the right subfigure demonstrates the effect of matched RRC filtering, where the constellation points are more tightly clustered, reflecting improved symbol separation and reduced inter-symbol interference. This visual evidence reinforces the quantitative findings from the MSE analysis, highlighting the critical role of pulse shaping in enhancing system performance.

What is particularly noteworthy is that the constellation diagram with RRC filtering exhibits a more compact and well-defined symbol distribution. This is because in the unfiltered scenario no upsampling takes place, and the symbols are transmitted at the symbol rate. Matlab measures the average power using Eq. (12). At 20dB, the noise variance is injected according to Eq. (13). The resulting standard deviation and average estimate are given by Eq. (6), Eq. (7), and Eq. (8).

$$\sigma_{\text{noise}} = \sqrt{2916} \approx 54 \quad (6)$$

$$\text{Estimated_Average} = \frac{\text{Ideal_Sum} + w}{K} = 27 + \frac{w}{20} \quad (7)$$

$$\sigma_{\text{Estimated_Average}} = \frac{\sigma_{\text{noise}}}{K} = \frac{54}{20} = 2.7 \quad (8)$$

In contrast, the RRC filtered scenario employs upsampling, so the RRC filter is normalized to unit energy Eq. (9). When the convolution happens with the upsampled signal, the energy is spread across the sps samples, resulting in a lower effective noise variance per sample. In greater detail, the noise variance after filtering is described by Eq. (10), and the standard deviation of the estimated average is given by Eq. (11).

$$\sum_n g[n]^2 = 1 \quad (9)$$

$$\sigma_{\text{noise, filtered}}^2 = \frac{\sigma_{\text{noise}}^2}{\text{sps}} = \frac{2916}{10} = 291.6 \quad (10)$$

$$\sigma_{\text{Estimated_Average, filtered}} = \frac{\sigma_{\text{noise, filtered}}}{K} = \frac{\sqrt{291.6}}{20} \approx 0.85 \quad (11)$$

$$P_{\text{signal}} = \frac{1}{N} \sum (\text{Ideal_Sum})^2 \approx 540^2 = 291600 \quad (12)$$

$$\sigma_{\text{noise}}^2 = \frac{P_{\text{signal}}}{100} = \frac{291600}{100} = 2916 \quad (13)$$

The eye diagram animation in Figure 4 illustrates the temporal evolution of the received signal's integrity as the number of samples per symbol varies. The left subfigure, corresponding to a truncated filter with span = 4, shows that as sps increases, the eye opening becomes wider, indicating improved timing margins and reduced inter-symbol interference. The right subfigure, with an extended filter of span = 12, demonstrates that even at lower sps values, the eye remains relatively open due to the longer filter length, which provides better pulse shaping and mitigates ISI effects. This visual evidence reinforces the quantitative findings from the MSE analysis, highlighting the critical role of both sps and filter span in enhancing system performance.

If we take a closer look near the top of each diagram 4 we see that the aggregated amplitude can take any real value, there are no voltage levels, and the eye diagram is not a square shape, but rather a continuous distribution of amplitudes. This is

because the transmitted signals are analog values, and the superposition of these signals results in a continuous range of possible amplitudes at the receiver. The eye diagram reflects this continuous nature, showing the variations in amplitude over time and the impact of noise on the received signal. The eye opening indicates the time intervals where the signal

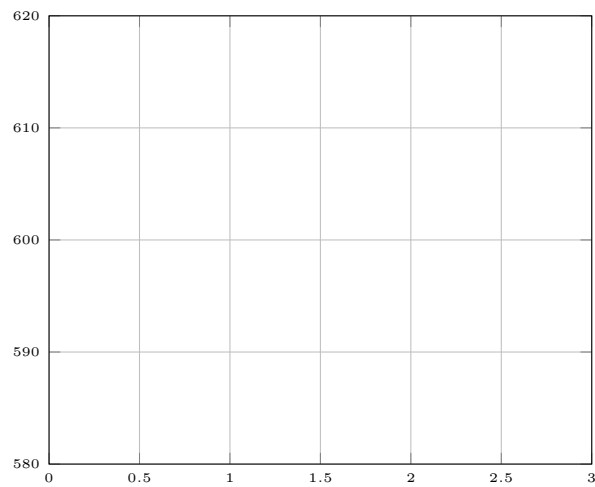


Figure 5: Simulink block diagram representing the ideal OTA computation network over an AWGN channel.

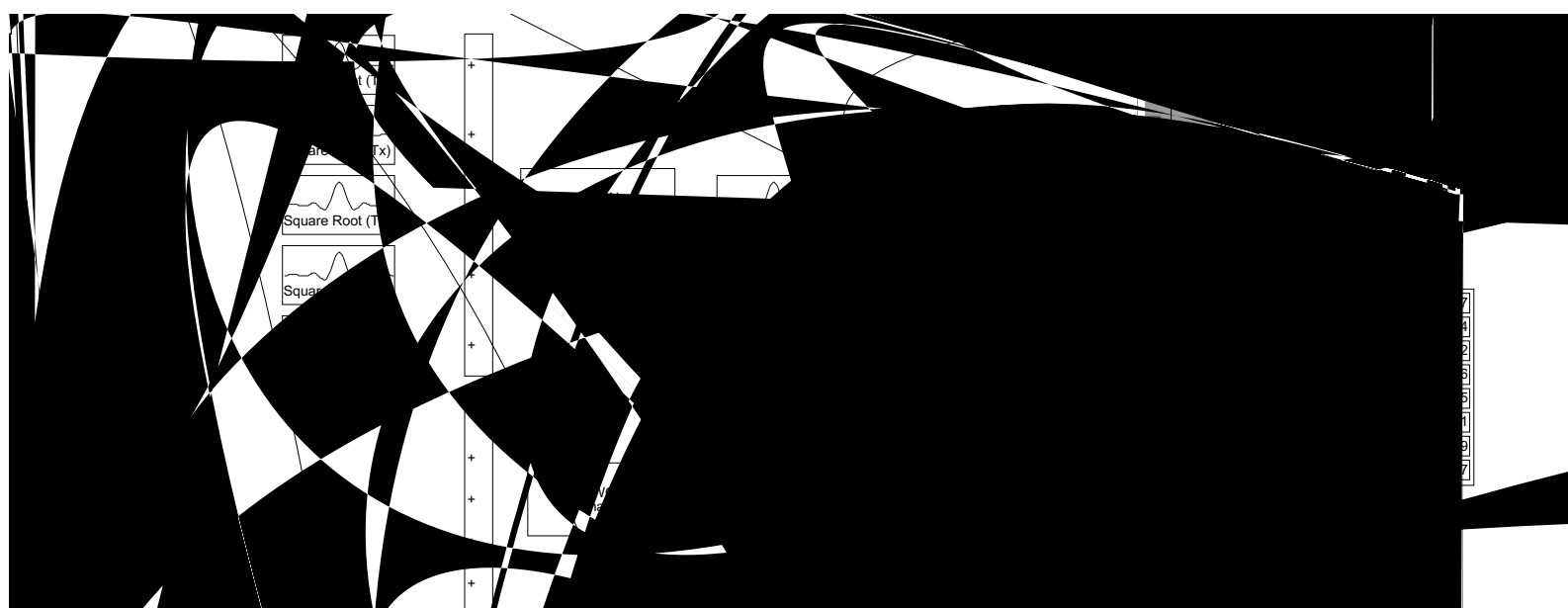


Figure 6: Simulink block diagram representing the ideal OTA computation network over an AWGN channel.

that practical over-the-air computation mandates time-varying transmission (as opposed to static constant-valued signals), as pulse shaping through Finite Impulse Response (FIR) filtering inherently requires symbol-rate sampling and subsequent interpolation. Consequently, systems employing matched filter pairs necessitate carefully synchronized symbol generation to avoid spurious transient responses.

The constellation diagrams provide a clear visual confirmation of the role played by pulse shaping in the received signal. The plot corresponding to the fusion center observation after the matched RRC filter shows a more compact and well-defined symbol distribution, indicating that the filtering stage has improved temporal alignment and reduced the impact of inter-symbol interference. In contrast, the constellation associated with the theoretical receiver model without pulse shaping appears more dispersed and less structured, reflecting the absence of the smoothing and energy-concentrating effect provided by the matched filter. This comparison highlights that the matched RRC filter contributes not only to spectral shaping, but also to preserving symbol integrity and improving the reliability of the recovered OTA estimate.

When an RRC filter is generated using MATLAB/Simulink's underlying `rcosdesign` math, the filter is normalized for Energy, not Peak Amplitude. The algorithm guarantees that the sum of the squares of the filter taps equals 1.0 ($\sum h[n]^2 = 1$). However, in discrete-time Digital Signal Processing (DSP), when you push a signal through an interpolation filter, the energy gets spread out across the sps samples. When the signal passes through the Tx RRC filter, travels through the air, and passes through the Rx RRC matched filter, the continuous-time math is perfect, but the discrete-time convolution sum inherently scales the output by a factor related to the sps.

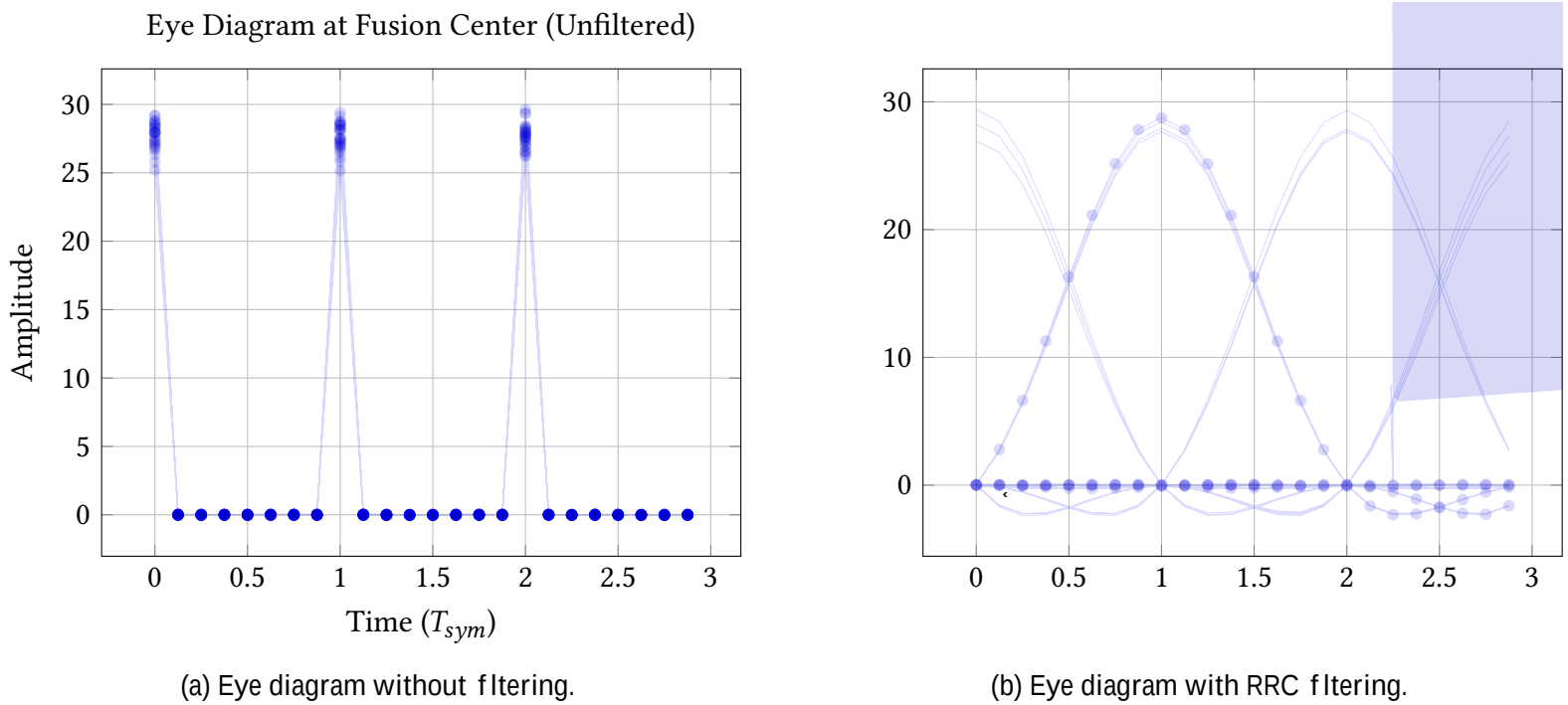


Figure 7: Eye Diagrams with and without filtering.

The eye diagrams further illustrate the impact of pulse shaping on the received signal. The eye diagram corresponding to the fusion center observation after the matched RRC filter shows a more open and well-defined eye pattern, indicating that the filtering stage has improved temporal alignment and reduced inter-symbol interference. In contrast, the eye diagram associated with the theoretical receiver model without pulse shaping appears more closed and less structured, reflecting the absence of the smoothing and energy-concentrating effect provided by the matched filter. This comparison highlights that the matched RRC filter contributes not only to spectral shaping, but also to preserving symbol integrity and improving the reliability of the recovered OTA estimate.

In Figure 8, the sensitivity of the Over-the-Air Computation system to symbol timing synchronization error is depicted. The plot illustrates that at the optimal strobe instant ($\tau_e = 0$), the Nyquist ISI cancellation effectively minimizes the error floor to 10^{-5} . However, even a slight clock drift of $0.25T_s$ results in a dramatic increase in error, exceeding seven orders of

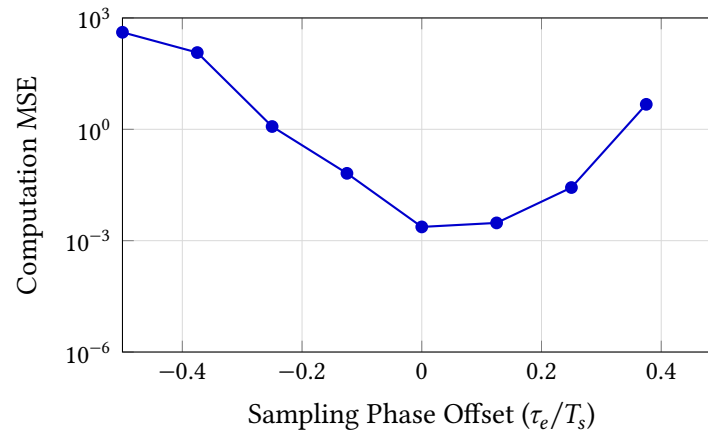


Figure 8: S-Curve of OTA computation to symbol timing synchronization error (sps = 8).

magnitude. This underscores the critical importance of precise timing synchronization in maintaining the integrity and accuracy of OTA computations, particularly in systems employing pulse shaping and matched filtering.

$$\text{MSE}(\phi) \quad \text{for } \phi \in \{0, 1, \dots, 7\} \quad (16)$$

$$\sum_{k=1}^K x_k(t - \tau_k) \quad (17)$$

$$\text{MSE}(\tau_e) = \frac{1}{N} \sum_{n=1}^N (\hat{x}(n, \tau_e) - x(n))^2 \quad (18)$$

1.d GNU Radio Validation

Finally, the ideal model is verified using a Software Defined Radio (SDR) environment in GNU Radio. To maintain the ideal conditions mathematically defined in (17), the Root Raised Cosine (RRC) filters are perfectly matched. Specifically, the interpolation rate at the transmitters and the decimation rate at the receiver are identically matched to the symbol design rate (**sps_actual = sps_design = 6**). In other words the upsampling rate of the FIR filters matches the spacing between sample generation.

An important architectural consideration emerges from the GNU Radio implementation: in the absence of explicit digital modulation schemes, strategic downsampling at the transmitter becomes mandatory to enable proper filter operation. Specifically, each individual sensor transmits one symbol while simultaneously discarding sixteen interpolated samples, thereby maintaining compliance with Nyquist sampling criterion while leveraging the spectral efficiency benefits of pulse shaping.

The matched filter perfectly recovered the symbol, but the amplitude fell short (26.4 instead of 27.4). The one-unit reduction in amplitude results from a fundamental property of Raised Cosine (RC) filters when calculating MSE. When a RRC filter is generated using MATLAB/Simulink's underlying `rcosdesign` math, the filter is normalized for Energy, not Peak Amplitude. The algorithm guarantees that the sum of the squares of the filter taps equals 1.0 ($\sum h[n]^2 = 1$). However, in discrete-time DSP, when a signal passes through an interpolation filter, the energy is spread out across the *SPS* samples. When the signal passes through the Tx RRC filter, travels through the air, and passes through the Rx RRC matched filter, the continuous-time math is perfect, but the discrete-time convolution sum inherently scales the output by a factor related to the sps.

In commercial transceivers, the filters are not expected to perfectly preserve the raw voltage scale. Instead, a static scalar multiplier (Gain block) is applied at the receiver, or an Automatic Gain Control (AGC) loop is used, to rescale the discrete

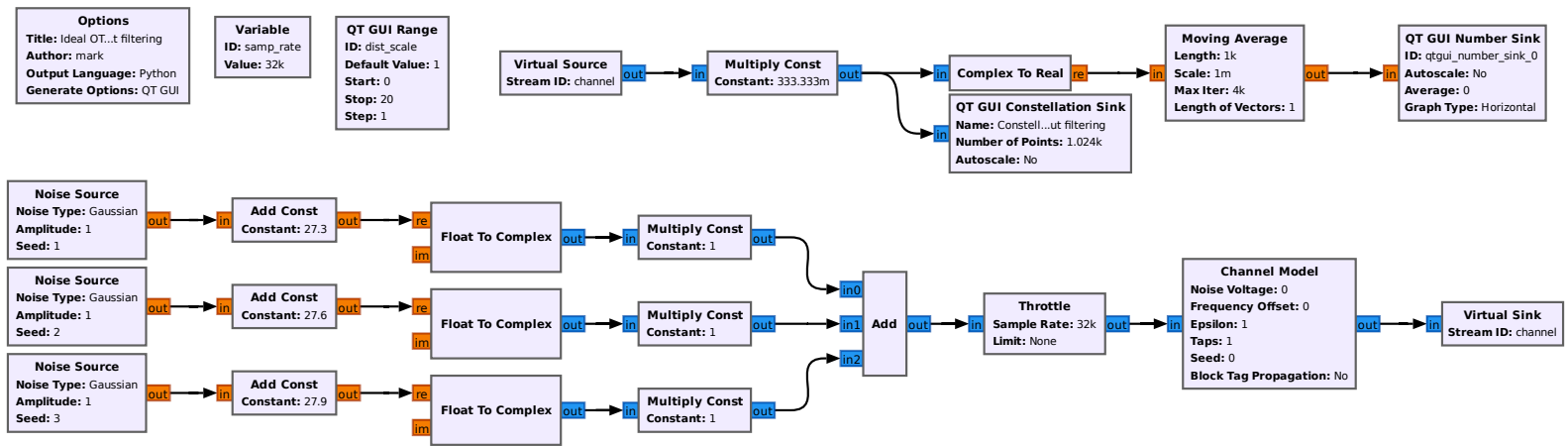


Figure 9: GNU Radio flowgraph for the ideal, Nyquist-compliant scenario.

digital values back to physical analogue units (such as Volts or Celsius) before the data reaches the Constellation Diagram or the Fusion Center average calculation.

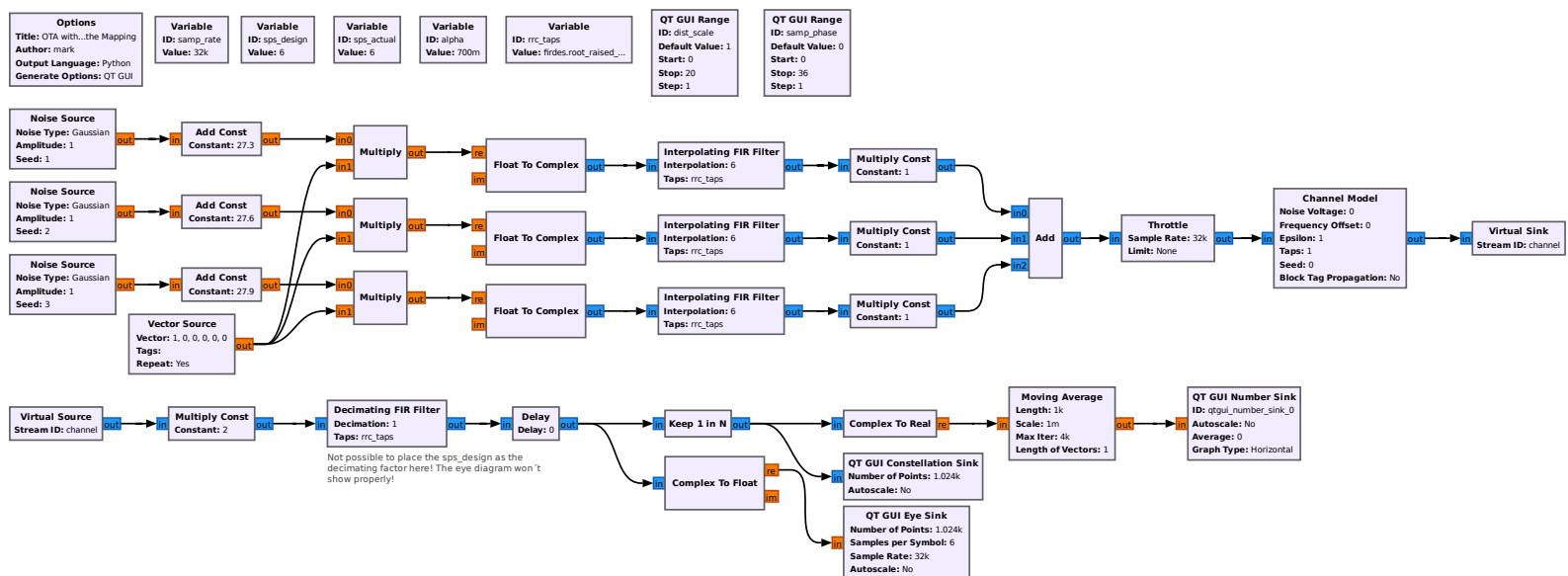


Figure 10: GNU Radio flowgraph for the ideal, Nyquist-compliant scenario.

Under these constraints, the Eye Diagram remains perfectly "open" at the optimal sampling phase, and the Constellation Diagram exhibits tightly clustered points. The absence of horizontal dispersion in the constellation confirms that, prior to the introduction of timing offsets or Faster-Than-Nyquist signaling, the MAC superposition principle flawlessly computes the arithmetic average of the agricultural sensors.

A critical implementation detail warranting emphasis concerns the optimal sampling phase selection. Empirical observation from the GNU Radio receiver indicates that maximum data recovery is achieved at a sampling phase offset of 33 samples out of the 96-sample cycle¹. This phase offset reflects the group delay inherent to the polyphase interpolating and decimating FIR filter implementations, necessitating careful calibration for optimal demodulation performance.

It is also useful to note that the GNU Radio implementation, while functionally equivalent to the MATLAB/Simulink model, introduces additional practical considerations, such as real-time processing constraints and hardware-induced latency. These factors may influence the observed performance metrics, particularly in scenarios where the system is subjected to dynamic channel conditions or when operating near the limits of the hardware's processing capabilities. Therefore, while the idealized models provide a valuable theoretical benchmark, real-world deployments must account for these additional complexities to ensure robust and reliable operation.

¹The fundamental cycle spans 16 symbols; with 6 samples per symbol, the total cycle length is $16 \times 6 = 96$ samples.

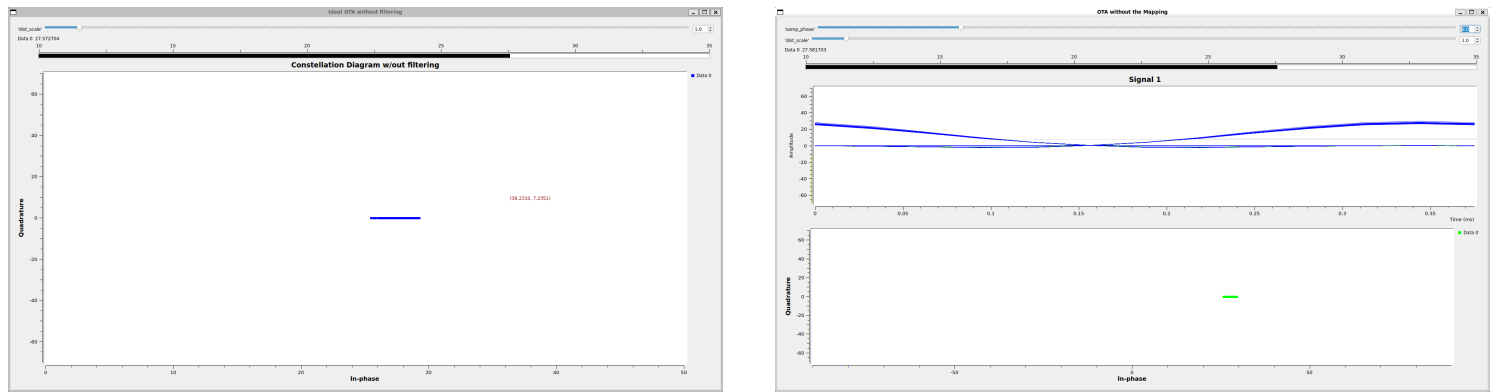


Figure 11: GNU Radio flowgraph for the ideal, Nyquist-compliant scenario.

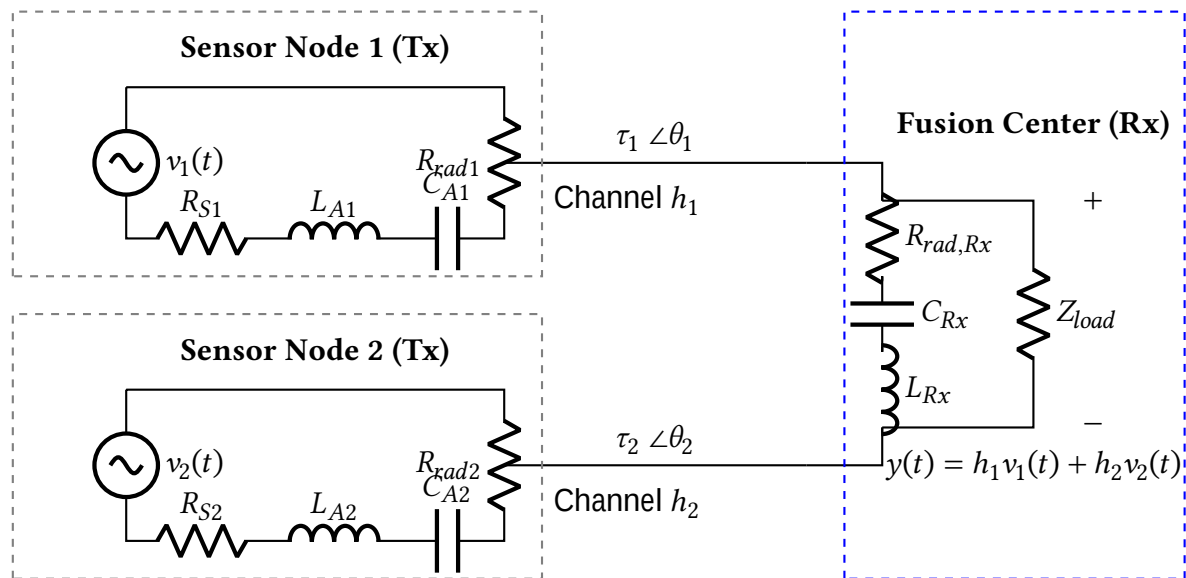


Figure 12: Circuit equivalent of the ideal OTA computation network. Each sensor node is modeled as a voltage source with internal impedance, and the fusion center aggregates the received signals through a common load. The wireless channels are represented as transmission lines with associated delays and phase shifts, capturing the effects of propagation in a simplified manner.

The circuit equivalent of the ideal OTA computation network is depicted in Figure 12. This representation abstracts the wireless channel into a simplified electrical model, where each sensor's transmission is treated as a voltage source, and the fusion center aggregates these signals through a common load. The circuit

References

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